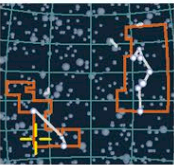


EMISSION NEBULA

Eagle Nebula



CATALOG NUMBER  
IC 4703

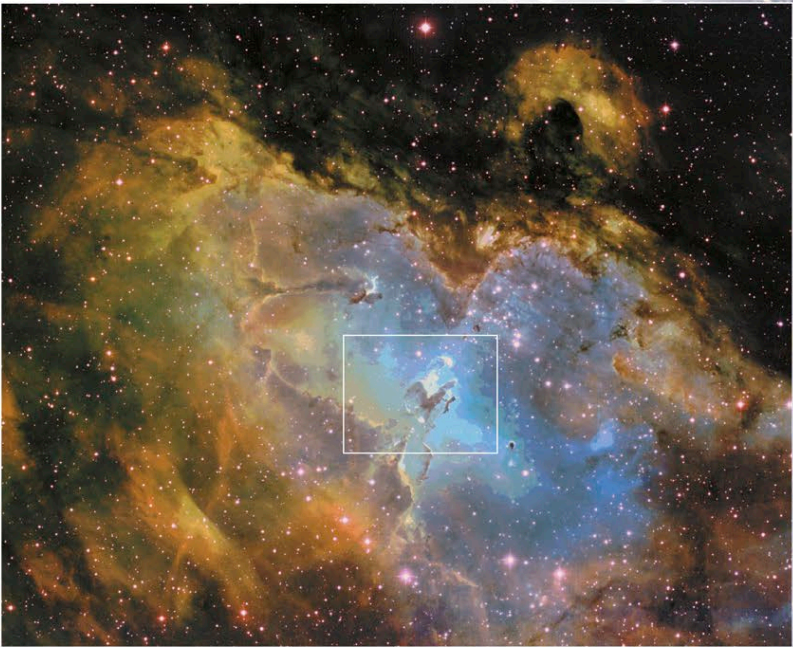
DISTANCE FROM SUN  
7,000 light-years

MAGNITUDE 6

SERPENS

Observations of the Eagle Nebula have introduced new ideas into the theory of star formation. Lying in one of the dense spiral arms of the Milky Way, this is an immense stellar nursery where young stars flourish, new stars are being created, and the material and triggers exist for future star formation. In optical wavelengths, this region is dominated by the light from the bright young star cluster M16. This cluster was discovered by the Swiss astronomer Philippe Loys de Chéseaux in around 1745, but it was nearly 20 years later that the surrounding nebula, from which the star cluster had formed, was discovered by Charles Messier (see p.73). The star cluster itself is only about 5 million years old and has a diameter of about 15 light-years. The Eagle Nebula is much larger than the star cluster, with a diameter of about 70 light-years.

In 1995, the Hubble Space Telescope imaged features within the nebula that are commonly known as the Pillars of Creation (see panel, below). These famous pillars are towers of dense material that have resisted evaporation by radiation from local young stars. However, the stars' ultraviolet radiation is gradually boiling their surfaces away through a process called photoevaporation. Because the towers themselves are not of a consistent density, the continuing photoevaporation has caused some of the smaller nodules, known as evaporating gaseous globules (EGGs), to become detached from the main gas towers. At this point, these dense stellar nurseries cease to accrue more material, and any embryonic star



HUGE STELLAR NURSERY

This wide-field image shows the immensity of the Eagle Nebula, with the three Pillars of Creation located near the center. This huge cloud of gas lies in the galaxy's Sagittarius–Carina arm, toward the galactic center.

within has its upper mass limit fixed. It is thought that this type of star formation inhibits the formation of accretion disks around the stars, which are believed to be the material from which planets are formed. These detailed images of the Pillars of Creation were the first to suggest this process of star creation. The Eagle Nebula also contains many Bok globules, regions where future star formation is probably occurring.

EXPLORING SPACE

THE PILLARS OF CREATION



This image, taken by the Hubble Space Telescope in 1995, has become one of the most famous and iconic astronomical images. Revealing for the first time in dramatic detail a previously unsuspected process of star formation, it captured the public's imagination and inspired a new interest in astronomy. The image's aesthetic appeal and the sense of wonder it inspires have led to it being displayed on posters, in magazines, and even on stamps.

STELLAR CLOSE-UP

These spectacular pillars of dust and gas are several light-years long but represent only a small section of the Eagle Nebula.